PROCEEDINGS OF THE
AMERICAN MATHEMATICAL SOCIETY
https://doi.org/10.1090/proc/16393
Article electronically published on June 23, 2023

SPHERICAL RANDOM SAMPLING OF LOCALIZED FUNCTIONS ON $\mathbb{S}^{n-1}$

DHIRAJ PATEL AND S. SIVANANTHAN

(Communicated by Yuan Xu)

Abstract. In this paper, we show that the set of random points uniformly distributed over $\mathbb{S}^{n-1}$ is a stable set of sampling for a class of localized functions in $L^2(\mathbb{S}^{n-1})$ with overwhelming probability. The probability bound of the random sampling inequality for the localized function is obtained from the matrix Bernstein inequality on the independent random matrices.

1. Introduction

Spherical sampling is a mathematical transfer technique of converting a continuous signal on a unit sphere to its samples at a preferable countable set so that the continuous signal can be recovered from its discrete samples without losing any information. The spherical sampling problem deals to find such a countable set on the sphere. The theory of spherical sampling is used extensively in many geoscience fields, such as geoexploration using gravitational and magnetic data, laser and radar technology, seismic tomography and reflection imaging, and many more. For numerous developments in spherical sampling and their applications in mathematical geodesy and geophysics, we refer the reader to [14].

In a sampling problem, the continuous functions are analyzed and recovered from their discrete sampling values. Hence, one needs to consider the function space in which the point evaluation of the considered functions should make sense [1]. In this paper, we consider a reproducing kernel subspace of square-integrable functions on a unit sphere $\mathbb{S}^{n-1} \subset \mathbb{R}^n$. Mathematically, the spherical sampling problem is stated as follows.

Given a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ of functions defined on $\mathbb{S}^{n-1}$, find a countable sampling set $\Gamma \subset \mathbb{S}^{n-1}$ such that

$$A\|f\|_{\mathcal{H}}^2 \leq \sum_{\gamma\in\Gamma} |f(\gamma)|^2 \leq B\|f\|_{\mathcal{H}}^2 \quad \forall f \in \mathcal{H}, \tag{1.1}$$

for some $A, B > 0$. This implies the sampling operator $f \mapsto (f(\gamma))_{\gamma\in\Gamma}$ is bounded from $\mathcal{H}$ to $\ell^2(\Gamma)$ and a small change of sample value $f(\gamma)$ leads to a small change of f. Hence, f can be stably reconstructed from its samples $\{f(\gamma) : \gamma \in \Gamma\}$ and the sampling set Γ is called *stable set of sampling* for $\mathcal{H}$. The set of $(2m+1)$-th root

Received by the editors August 12, 2022, and, in revised form, December 18, 2022.

2020 *Mathematics Subject Classification.* Primary 94A20, 41A17, 42C15, 60G50, 60G99; Secondary 60B20.

The first author was financially assisted by the Council of Scientific and Industrial Research, New Delhi. The second author acknowledges the financial support of the Science and Engineering Research Board, Government of India, through project no. CRG/2019/002412.

of unity is a stable sample set for the space of m-degree trigonometric polynomials, and the upper and lower bounds of sampling inequality (1.1) are independent of dimension of the space [39]. The sampling inequality of this type is known as Marcinkiewicz-Zygmund inequality and the collection of all such stable sample sets is called the Marcinkiewicz-Zygmund family. The Marcinkiewicz-Zygmund inequality was generalized to the space of spherical harmonic polynomials [6, 13, 23, 26, 27]. In recent years, the problem was studied for Sobolev space [17], Bergman space [18], finite-dimensional function space on a compact set [35], and so on. The Marcinkiewicz-Zygmund family can be used to derive the stable sample set for infinite-dimensional RKHS [16, 18]. However, for the Hardy space on the unit disk, it was shown that there does not exist any stable sample set [36]. Note that for an infinite-dimensional RKHS defined on a compact set, a stable sample set must contain infinitely many points and the sampling set should have an accumulated point. This may lead to unboundedness of the sampling operator. Therefore, the study of sampling problem on an infinite-dimensional subspace of $L^2(\mathbb{S}^{n-1})$ is challenging. In this paper, we consider an infinite-dimensional RKHS $\mathcal{H}$ and estimate the probability of randomly chosen finitely many sample points from $\mathbb{S}^{n-1}$ to be a stable sample set for a certain class of functions V in $\mathcal{H}$. In other words, we established that any f in V is determined uniquely and stably from its random sample values with high probability. As a consequence, we can employ exponential convergent iterative algorithms to construct f from its random samples (see [24, 28]). In contrast to that in a seminal work of Cucker and Smale [9] discussed the problem of recovery of a function from random samples in the framework of learning theory. Further, Smale and Zhou [31, 32] discussed the connection between sampling sets and function reconstruction from their sample values. Recently, a significant study on the worst-case error analysis of sampling recovery of function from their random sample values has been addressed in [11, 21, 22, 37].

In many applications in the fields of image processing [8], learning theory [9, 30], and compressed sensing [12], functions are recovered from their random measurements. Candés, Romberg and Tao [7] studied the reconstruction of a sparse trigonometric polynomial from random frequency samples. Recently, the random sampling problem was studied for the subspace of Lebesgue space $L^p(\mathbb{R}^n)$. For example, Paley-Wiener space, shift-invariant space, continuous function space with bounded derivative, and image space of the idempotent integral operator [4, 5, 15, 24, 29, 38]. The random sampling problem for the space of spherical harmonic polynomial of finite degree was studied by Bass and Gröchenig [3].

In this paper, we study the spherical random sampling problem on an infinite-dimensional RKHS $\mathcal{H}$ (Section 2) which is continuously embedded in $L^2(\mathbb{S}^{n-1})$. We consider the set $\mathcal{H}(\delta)$ of δ-concentrated functions with respect to $\|\cdot\|_{L^2(\mathbb{S}^{n-1})}$ in $\mathcal{H}$, i.e., for $\delta > 0$,

$$\mathcal{H}(\delta) = \big\{ f \in \mathcal{H} : \|f\|^2_{L^2(\mathbb{S}^{n-1})} \geq (1-\delta)\|f\|^2_{\mathcal{H}} \big\}.$$

We derive the probability estimate of the random sampling inequality for $\mathcal{H}(\delta)$. Since the set $\mathcal{H}(\delta)$ contains finite-dimensional space of spherical harmonic polynomials, the result generalizes the classical random sampling problem for finite-dimensional space [3]. In Section 3, we note that the set $\mathcal{H}(\delta)$ is not contained in a finite-dimensional space. To the best of our knowledge, the random sampling problem for the class of functions in an infinite-dimensional RKHS $\mathcal{H}$ on $\mathbb{S}^{n-1}$ has not been explored in the literature.

The main theorem of this paper is stated as follows.

Theorem 1.1. *Let $\{x_i : i \in \mathbb{N}\}$ be a sequence of i.i.d. random variables which are uniformly distributed over $\mathbb{S}^{n-1}$ and ω_{n-1} denotes the surface area of $\mathbb{S}^{n-1}$. Then for $\epsilon \in (0,1)$, the sampling inequality*

$$\left(\frac{1-2\sqrt{\delta}}{\omega_{n-1}} - 2D^2\sqrt{2\delta} - \nu\right)\|f\|^2_{L^2(\mathbb{S}^{n-1})} \le \frac{1}{r}\sum_{i=1}^{r}|f(x_i)|^2 \le \frac{D^2}{(1-\delta)}\|f\|^2_{L^2(\mathbb{S}^{n-1})} \tag{1.2}$$

holds for all $f \in \mathcal{H}(\delta)$ with probability at least $1-\epsilon$ if

$$r > \frac{D^2(3+\nu\omega_{n-1})}{\nu^2\omega_{n-1}} \log \frac{2^{n-1}(n+1)\left(1+\left(\frac{2(n-2)!}{C}\right)^{\frac{n-1}{n+\alpha-2}}\right)}{\epsilon}.$$

Here the constants $C > 0$, D depends on n, $\alpha > 1$, δ and ν satisfy the following inequality

$$\delta < \frac{1}{4(1+\sqrt{2}D^2\omega_{n-1})^2} \quad \text{and} \quad 0 < \nu < \frac{1-2\sqrt{\delta}}{\omega_{n-1}} - 2D^2\sqrt{2\delta}.$$

The paper is organized as follows. In Section 2, we give some preliminary results and examples of RKHS. In Section 3, we estimate the error bound of the approximation of δ-concentrated function with the spherical harmonic polynomials and we establish the sampling inequality for a class of δ-concentrated functions. In Section 4, we define independent random variables with respect to uniformly distributed random samples over $\mathbb{S}^{n-1}$, and then using the probabilistic estimate for the matrix norm of the sum of random variables, we prove the main Theorem 1.1. Further, we discuss the extension of the main result in a more general RKHS.

2. Preliminaries

In this section, we define RKHS of functions on $\mathbb{S}^{n-1}$ which is continuously embedded in $L^2(\mathbb{S}^{n-1})$ and give some examples.

It is well known that functions in $L^2(\mathbb{S}^{n-1})$ can be represented in terms of spherical harmonic functions [10]. In particular, $L^2(\mathbb{S}^{n-1}) = \bigoplus_{k=0}^{\infty} H_k$, where H_k is the space of spherical harmonic polynomial of degree k and dimension d_k. The spherical harmonic representation of $f \in L^2(\mathbb{S}^{n-1})$ is given by $f = \sum\limits_{k=0}^{\infty}\sum\limits_{j=1}^{d_k} c_{j,k}Y_{j,k}$, where $\{Y_{j,k} : j = 1, \ldots, d_k\}$ is the orthonormal basis of H_k. However, the point evaluation functionals on $L^2(\mathbb{S}^{n-1})$ are not continuous.

A Hilbert space $\mathcal{H}$ of functions on $\mathbb{S}^{n-1}$ is a reproducing kernel Hilbert space if for each $x \in \mathbb{S}^{n-1}$ the point evaluation functional $f \mapsto f(x)$ is continuous, i.e., for each $x \in \mathbb{S}^{n-1}$ there exists a constant $C_x > 0$ such that

$$|f(x)| \le C_x\|f\|_{\mathcal{H}}, \quad \forall f \in \mathcal{H}.$$

From Lemma 4.2 and Theorem 4.20 in [34], the following result provides a characterization for RKHS.

Theorem 2.1. *Let $\mathcal{H}$ be a separable Hilbert space of functions over X with orthonormal basis $\{\phi_k\}_{k\in\mathbb{Z}}$. Then $\mathcal{H}$ is a RKHS if and only if $\big(\phi_k(x)\big)_{k\in\mathbb{Z}} \in \ell^2(\mathbb{Z})$ for all $x \in X$. Moreover, the reproducing kernel is uniquely defined by*

$$K(x,y) = \sum_{k\in\mathbb{Z}} \overline{\phi_k(x)}\phi_k(y).$$

The above result does not ensure the boundedness of a function in RKHS $\mathcal{H}$. If the reproducing kernel K of $\mathcal{H}$ is bounded on X, i.e., $\sup_{x\in X} K(x,x) < \infty$, then every f in $\mathcal{H}$ is bounded and the converse is also true, see [34, Lemma 4.23].

The collection $\{Y_{j,k} : j = 1, \dots, d_k, k \in \mathbb{N}_0\}$ is an orthonormal basis of $L^2(\mathbb{S}^{n-1})$, where $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. If ω_{n-1} is the surface area of $\mathbb{S}^{n-1}$, then for each $k \geq 0$,

$$\sum_{j=1}^{d_k} |Y_{j,k}(x)|^2 = \frac{d_k}{\omega_{n-1}}, \tag{2.1}$$

for all $x \in \mathbb{S}^{n-1}$ [33]. This implies $L^2(\mathbb{S}^{n-1})$ is not a RKHS. Consequently, to be a reproducing kernel Hilbert subspace of $L^2(\mathbb{S}^{n-1})$, we need to assume some additional condition on the decay of the spherical harmonic coefficients. In this paper, we consider the Hilbert space as follows:

$$\mathcal{H} = \mathcal{H}_\Psi := \left\{ f = \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k} Y_{j,k} \in L^2(\mathbb{S}^{n-1}) : \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} |c_{j,k}|^2 \psi_k < \infty \right\},$$

where $\Psi = \{\psi_k : k \in \mathbb{N}_0\}$ is a sequence of positive real, and for $f, g \in \mathcal{H}_\Psi$ the inner product is defined by

$$\langle f, g\rangle_{\mathcal{H}_\Psi} = \left\langle \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k} Y_{j,k}, \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} d_{j,k} Y_{j,k} \right\rangle_{\mathcal{H}_\Psi} := \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k}\overline{d_{j,k}}\psi_k.$$

If $\psi_k \neq 0$ for finitely many $k \in \mathbb{N}_0$ then $\mathcal{H}_\Psi$ is a space of band-limited functions on $\mathbb{S}^{n-1}$ [3, 20]. The space $\mathcal{H}_\Psi$ is a separable Hilbert space with an orthonormal basis $\left\{\frac{1}{\sqrt{\psi_k}} Y_{j,k} : 1 \leq j \leq d_k, k \in \mathbb{N}_0\right\}$. Hence, by Theorem 2.1, $\mathcal{H}_\Psi$ is a RKHS if and only if

$$\sum_{k=0}^{\infty}\sum_{j=1}^{d_k} \frac{1}{\psi_k}|Y_{j,k}(x)|^2 = \frac{1}{\omega_{n-1}}\sum_{k=0}^{\infty}\frac{d_k}{\psi_k} < \infty.$$

This implies, ψ_k must diverge to ∞ at the rate of at least $o(d_k k)$. Hence, we assume that $\{\psi_k\}_{k=0}^{\infty}$ is a non-decreasing sequence with $\psi_0 = 1$ and satisfies

$$\psi_k > C d_k k^{\alpha}, \quad \forall\ k \geq 1, \tag{2.2}$$

for some $C > 0$ and $\alpha > 1$.

Corollary 2.2. *If $\{\psi_k\}_{k=0}^{\infty}$ is a sequence satisfying (2.2), then $\mathcal{H}_\Psi$ is RKHS of bounded functions. Moreover, there exists $D = \frac{1}{\sqrt{\omega_{n-1}}}\left(1 + \frac{1}{C(\alpha-1)}\right)^{\frac{1}{2}}$ such that for each $f \in \mathcal{H}_\Psi$,*

$$\|f\|_{L^\infty(\mathbb{S}^{n-1})} \leq D\|f\|_{\mathcal{H}_\Psi}.$$

Proof. For $x \in \mathbb{S}^{n-1}$, equality (2.1) and inequality (2.2) imply

$$\sum_{k=0}^{\infty}\sum_{j=1}^{d_k} \frac{1}{\psi_k}|Y_{j,k}(x)|^2 \leq D^2. \tag{2.3}$$

Therefore, spherical harmonic representation of $f \in \mathcal{H}_\Psi$ and Hölder's inequality imply

$$|f(x)| \leq D\|f\|_{\mathcal{H}_\Psi}, \qquad \text{for all } f \in \mathcal{H}_\Psi.$$

This completes the proof. $\square$

For more details about spherical harmonics and its interesting properties, we refer the reader to [10]. Now we discuss a few examples of well-known RKHS with appropriate choice of $\Psi = \{\psi_k\}_{k=0}^{\infty}$.

Example 2.1. The Sobolev space $H^s(\mathbb{S}^{n-1})$, $s > (n-1)/2$ is a space of all continuous functions f in $L^2(\mathbb{S}^{n-1})$ such that $(-\Delta_S)^{\frac{s}{2}} f$ in $L^2(\mathbb{S}^{n-1})$, where Δ_S is the Laplace-Beltrami operator on $\mathbb{S}^{n-1}$. The space $H^s(\mathbb{S}^{n-1})$ is a Hilbert space with the norm

$$\|f\|^2_{H^s(\mathbb{S}^{n-1})} = \|f\|^2_{L^2(\mathbb{S}^{n-1})} + \|(-\Delta_S)^{\frac{s}{2}} f\|^2_{L^2(\mathbb{S}^{n-1})}.$$

Moreover, the spherical harmonic polynomials $Y_{j,k}$ are eigenfunctions of $(-\Delta_S)$ corresponding to the eigenvalue $k(k+n-2)$ for each $1 \leq j \leq d_k$ (cf. [10]). Then the Hilbert space norm of $H^s(\mathbb{S}^{n-1})$ can be rewritten as

$$\|f\|^2_{H^s(\mathbb{S}^{n-1})} = \sum_{k=0}^{\infty} \sum_{j=1}^{d_k} \big(1 + k^s(k+n-2)^s\big) |c_{j,k}|^2, \tag{2.4}$$

where $c_{j,k}$ is the spherical harmonic coefficient of f. Hence, for $\psi_k = \big(1 + k^s(k + n-2)^s\big)$ the space $\mathcal{H}_\Psi$ describes the Sobolev space $H^s(\mathbb{S}^{n-1})$.

Example 2.2. Let $F : [-1,1] \to \mathbb{C}$ be a continuous function. Then the Funk-Hecke Formula [19] states that for each j, $1 \leq j \leq d_k$, and $x \in \mathbb{S}^{n-1}$,

$$\int_{\mathbb{S}^{n-1}} F(x \cdot y) Y_{j,k}(y)\, d\sigma_n(y) = \lambda_k Y_{j,k}(x),$$

where σ_n denotes the surface measure on the sphere and

$$\lambda_k = \omega_{n-1} \int_{-1}^{1} F(t) P_k^n(t) (1-t^2)^{\frac{n-3}{2}}\, dt,$$

P_k^n is the Legendre polynomial of degree k in dimension n, and for $x, y \in \mathbb{R}^n$, $x \cdot y = \sum_{i=1}^n x_i y_i$.

If $F(t) = \sum\limits_{k=0}^{\infty} b_k t^k$ with $\sum\limits_{k=0}^{\infty} |b_k| < \infty$, then Azevedo and Menegatto [2] proved that there exists $M > 0$ such that

$$\lambda_k \leq \frac{M b_k}{2^{k+1} k^{\frac{n-2}{2}}}, \quad k \geq 1.$$

Put $\psi_k = \lambda_k^{-1}$, then $\mathcal{H}_\Psi$ is a RKHS with the reproducing kernel $F(x \cdot y)$. In particular, for $x, y \in \mathbb{S}^{n-1}$, $F(x \cdot y) = e^2 \exp(-\|x - y\|^2)$ can be rewritten as

$$F(x \cdot y) = \sum_{k=0}^{\infty} \frac{2^k}{k!} \langle x, y \rangle^k.$$

Hence the Hilbert space $\mathcal{H}_\Psi$ represents the Gaussian RKHS.

3. Sampling inequality for localized function on $\mathbb{S}^{n-1}$

In this section, we introduce a notion of localized (δ-concentrated) functions on $\mathbb{S}^{n-1}$ and discuss its approximation via the spherical harmonic polynomials.

Definition 3.1. A function f in $\mathcal{H}_\Psi$ is said to be δ-*concentrated* with respect to $\|\cdot\|_{L^2(\mathbb{S}^{n-1})}$ if

$$\|f\|^2_{L^2(\mathbb{S}^{n-1})} \geq (1-\delta) \|f\|^2_{\mathcal{H}_\Psi}.$$

The set $\mathcal{H}_\Psi(\delta)$ is a collection of all δ-concentrated functions in $\mathcal{H}_\Psi$.

Intuitively, the set $\mathcal{H}_\Psi(\delta)$ describes certain localization in $L^2(\mathbb{S}^{n-1})$ of smooth functions in $\mathcal{H}_\Psi$ in terms of spherical harmonic coefficients. It is easy to see that for any f in $\mathcal{H}_\Psi$, there exists a δ in $(0,1)$ such that f in $\mathcal{H}_\Psi(\delta)$. On the other hand, given $\delta > 0$, choose N such that $1 - \frac{1}{\psi_N} < \delta$, then

$$(1-\delta)\sum_{k=0}^{N}\sum_{j=1}^{d_k}|c_{j,k}|^2\psi_k \leq (1-\delta)\psi_N\sum_{k=0}^{N}\sum_{j=1}^{d_k}|c_{j,k}|^2 \leq \sum_{k=0}^{N}\sum_{j=1}^{d_k}|c_{j,k}|^2.$$

This implies, $\mathcal{H}_\Psi(\delta)$ contains the space of spherical harmonic polynomials of degree at most N. However, the set $\mathcal{H}_\Psi(\delta)$ is not contained in a finite-dimensional space. For example, consider $\psi_0 = 1$ and for $k \geq 1$, $\psi_k = d_k k^\alpha$, where $\alpha > 1$. For fixed $\delta \in (0,1)$, there exists $\beta > 0$ such that

$$(1-\delta)\sum_{k=1}^{\infty}\frac{1}{k^{\beta+1}} \leq \sum_{k=1}^{\infty}\frac{1}{k^{n+\alpha+\beta}}.$$

Then $\left\{f_N := \displaystyle\sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k}^N Y_{j,k} \,:\, N \in \mathbb{N}\right\}$ is a linearly independent set in $\mathcal{H}_\Psi(\delta)$, where the spherical harmonic coefficient $c_{j,k}^N$ is defined as follows:

$$c_{j,k}^N = \begin{cases} 1, & k=0,\, j=0, \\ 0, & k=N,\, 1\leq j < d_k, \\ \frac{1}{\sqrt{k^{n+\alpha+\beta}}}, & k = N,\, j = d_k, \\ \frac{1}{\sqrt{d_k k^{n+\alpha+\beta}}}, & k \neq N,\, 1 \leq j \leq d_k. \end{cases}$$

In the following, we estimate the error bound of the difference between the δ-concentrated function and its projection on the space of spherical harmonic polynomials. Let P_m be the orthogonal projection from $\mathcal{H}_\Psi$ to A_m (the space of spherical harmonic polynomials of degree at most m), and it is defined by

$$P_m f = P_m\Big(\sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k}Y_{j,k}\Big) = \sum_{k=0}^{m}\sum_{j=1}^{d_k} c_{j,k}Y_{j,k}.$$

Lemma 3.2. *If $f \in \mathcal{H}_\Psi(\delta)$ and $m \in \mathbb{N}$ is such that $\psi_m > 1$, then*

(i) $\|P_m f\|^2_{\mathcal{H}_\Psi} \geq \|P_m f\|^2_{L^2(\mathbb{S}^{n-1})} \geq \Big(1 - \frac{\psi_m\delta}{\psi_m - 1}\Big)\|f\|^2_{\mathcal{H}_\Psi}$,
(ii) $\|f - P_m f\|^2_{\mathcal{H}_\Psi} \leq \frac{\psi_m\delta}{\psi_m-1}\|f\|^2_{\mathcal{H}_\Psi}$,
(iii) $\|f - P_m f\|^2_{L^2(\mathbb{S}^{n-1})} \leq \frac{1}{\psi_m}\|f\|^2_{\mathcal{H}_\Psi}$.

Proof. Let $f = \sum_{k=0}^{\infty}\sum_{j=1}^{d_k} c_{j,k}Y_{j,k} \in \mathcal{H}_\Psi(\delta)$. As the collection $\{Y_{j,k} : 1 \le j \le d_k, k \in \mathbb{N}_0\}$ is an orthonormal basis of $L^2(\mathbb{S}^{n-1})$, we have

$$\begin{aligned}\|f\|^2_{\mathcal{H}_\Psi} - \|f - P_m f\|^2_{\mathcal{H}_\Psi} &\ge \sum_{k=0}^{m}\sum_{j=1}^{d_k}|c_{j,k}|^2 \\ &= \|f\|^2_{L^2(\mathbb{S}^{n-1})} - \sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2 \\ &\ge (1-\delta)\|f\|^2_{\mathcal{H}_\Psi} - \frac{1}{\psi_m}\sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2\psi_k,\end{aligned}$$

which further implies

$$\psi_m\delta\|f\|^2_{\mathcal{H}_\Psi} \ge \Big(\psi_m - 1\Big)\sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2\psi_k.$$

(i) Now, from the above inequality

$$\begin{aligned}\|P_m f\|^2_{L^2(\mathbb{S}^{n-1})} &= \sum_{k=0}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2 - \sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2 \\ &\ge \|f\|^2_{L^2(\mathbb{S}^{n-1})} - \frac{1}{\psi_m}\sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2\psi_k \\ &\ge (1-\delta)\|f\|^2_{\mathcal{H}_\Psi} - \frac{1}{\psi_m}\frac{\psi_m\delta}{\psi_m - 1}\|f\|^2_{\mathcal{H}_\Psi} \\ &= \Big(1 - \frac{\psi_m\delta}{\psi_m - 1}\Big)\|f\|^2_{\mathcal{H}_\Psi}.\end{aligned}$$

(ii) The bound of $\|f - P_m f\|^2_{\mathcal{H}_\Psi}$ follows immediately from the relations $\|f - P_m f\|^2_{\mathcal{H}_\Psi} = \|f\|^2_{\mathcal{H}_\Psi} - \|P_m f\|^2_{\mathcal{H}_\Psi}$ and $\|P_m f\|^2_{L^2(\mathbb{S}^{n-1})} \le \|P_m f\|^2_{\mathcal{H}_\Psi}$.

(iii) As $\{\psi_k\}$ is a non-decreasing sequence, we have

$$\|f - P_m f\|^2_{L^2(\mathbb{S}^{n-1})} \le \frac{1}{\psi_m}\sum_{k=m+1}^{\infty}\sum_{j=1}^{d_k}|c_{j,k}|^2\psi_k \le \frac{1}{\psi_m}\|f\|^2_{\mathcal{H}_\Psi}.$$

This completes the proof. □

In Lemma 3.3, we show that the error estimation of sample energy and square integral norm of functions in A_m imply the stability of sampling set for $\mathcal{H}_\Psi(\delta)$.

Lemma 3.3. *Let σ_n be the surface measure of $\mathbb{S}^{n-1}$ and D be the constant defined in Corollary 2.2. Let $X = \{x_i : i = 1, 2, \dots, r\}$ be a finite subset of $\mathbb{S}^{n-1}$ and $m \in \mathbb{N}$ be such that $\psi_m > 1$. If $\nu > 0$ and for all $p \in A_m$*

$$\frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}|p(x)|^2\,d\sigma_n(x) - \frac{1}{r}\sum_{i=1}^{r}|p(x_i)|^2 \le \nu\|p\|^2_{\mathcal{H}_\Psi}, \tag{3.1}$$

then

$$A(m)\|f\|^2_{L^2(\mathbb{S}^{n-1})} \le \frac{1}{r}\sum_{i=1}^{r}|f(x_i)|^2 \le \frac{D^2}{(1-\delta)}\|f\|^2_{L^2(\mathbb{S}^{n-1})} \quad \forall f \in \mathcal{H}_\Psi(\delta), \tag{3.2}$$

where $A(m) = \frac{1}{\omega_{n-1}}\Big(1 - \frac{\psi_m\delta}{\psi_m - 1}\Big) - 2D^2\sqrt{\frac{\psi_m\delta}{\psi_m - 1}} - \nu$.

Proof. For $f \in \mathcal{H}_\Psi(\delta)$, the upper bound of the sampling inequality (3.2) is easy to prove. To prove the lower bound, we consider the orthogonal decomposition $f = P_m f + (f - P_m f)$ of f. Then the triangle inequality implies

$$\begin{aligned}
&\big\|\big(f(x_i)\big)\big\|_{\ell^2(X)} \geq \big\|\big(P_m f(x_i)\big)\big\|_{\ell^2(X)} - \big\|\big((f-P_m f)(x_i)\big)\big\|_{\ell^2(X)}\\
&\quad\sum_{i=1}^{r}|f(x_i)|^2 \geq \sum_{i=1}^{r}|P_m f(x_i)|^2 - 2\big\|\big(P_m f(x_i)\big)\big\|_{\ell^2(X)}\big\|\big((f-P_m f)(x_i)\big)\big\|_{\ell^2(X)}\\
&\frac{1}{r}\sum_{i=1}^{r}|f(x_i)|^2 \geq \frac{1}{r}\sum_{i=1}^{r}|P_m f(x_i)|^2 - 2D^2\|P_m f\|_{\mathcal{H}_\Psi}\|f - P_m f\|_{\mathcal{H}_\Psi},
\end{aligned}$$

where the last step follows from Corollary 2.2. Using Lemma 3.2 and inequality (3.1), we have

$$\begin{aligned}
\frac{1}{r}\sum_{i=1}^{r}|f(x_i)|^2 &\geq \frac{1}{r}\sum_{i=1}^{r}|P_m f(x_i)|^2 - 2D^2\sqrt{\frac{\psi_m\delta}{\psi_m - 1}}\|f\|^2_{\mathcal{H}_\Psi}\\
&\geq \frac{1}{\omega_{n-1}}\|P_m f\|^2_{L^2(\mathbb{S}^{n-1})} - \nu\|P_m f\|^2_{\mathcal{H}_\Psi} - 2D^2\sqrt{\frac{\psi_m\delta}{\psi_m - 1}}\|f\|^2_{\mathcal{H}_\Psi}\\
&\geq \left[\frac{1}{\omega_{n-1}}\Big(1 - \frac{\psi_m\delta}{\psi_m - 1}\Big) - 2D^2\sqrt{\frac{\psi_m\delta}{\psi_m - 1}} - \nu\right]\|f\|^2_{L^2(\mathbb{S}^{n-1})}.
\end{aligned}$$

This completes the proof. □

Remark 3.4. The lower bound of the sampling inequality (3.2) is non-decreasing in m, and depends on ν and δ. For a given $\nu, \delta > 0$, there exists $m_0 \in \mathbb{N}$ such that $A(m) > 0$ for all $m \geq m_0 + 1$. In particular, if $0 < \nu < \frac{(1-2\sqrt{\delta})}{\omega_{n-1}} - 2D^2\sqrt{2\delta}$ and $\delta < \frac{1}{4(1+\sqrt{2}D^2\omega_{n-1})^2}$, then for all $m \geq m_0 + 1$,

$$A(m) \geq \frac{(1 - 2\sqrt{\delta})}{\omega_{n-1}} - 2D^2\sqrt{2\delta} - \nu =: A > 0,$$

where $m_0 = \max\{k \in \mathbb{N}_0 : \psi_k \leq 2\}$. In addition, if the sampling set $\{x_i : i = 1, \ldots, r\}$ on $\mathbb{S}^{n-1}$ satisfies (3.1) for A_m, where $m \geq m_0 + 1$, then for all $f \in \mathcal{H}_\Psi(\delta)$,

$$A\|f\|^2_{L^2(\mathbb{S}^{n-1})} \leq \frac{1}{r}\sum_{i=1}^{r}|f(x_i)|^2 \leq \frac{D^2}{(1-\delta)}\|f\|^2_{L^2(\mathbb{S}^{n-1})}.$$

Since $\frac{(m+1)^{n-2}}{(n-2)!} \leq d_m \leq m^{n-1}$ and ψ_m satisfy (2.2) for each $m \in \mathbb{N}_0$, we have $\psi_m > 2$ for all $m > \left[\frac{2(n-2)!}{C}\right]^{\frac{1}{n+\alpha-2}}$. This implies $m_0 \leq \left[\frac{2(n-2)!}{C}\right]^{\frac{1}{n+\alpha-2}}$.

4. Spherical random sampling

In the following, we consider uniformly distributed random samples from $\mathbb{S}^{n-1}$ and estimate the probability for the sampling inequality for $\mathcal{H}_\Psi(\delta)$. Lemma 3.3 provides the condition for which the sampling set is stable for $\mathcal{H}_\Psi(\delta)$. Hence the random spherical sampling problem reduces to estimate the probability for inequality (3.1).

For the convenience, we rearrange $\{Y_{j,k}\}$ and $\{\psi_k\}$ as $\{\phi_k\}$ and $\{\lambda_k\}$ respectively in the following way. For $k=1$,

$$\lambda_1 = \frac{1}{\psi_0} \quad \text{and} \quad \phi_1 = Y_{0,0},$$

and for $\dim(A_{m-1}) < k \leq \dim(A_m)$,

$$\lambda_k = \frac{1}{\psi_m} \quad \text{and} \quad \phi_k = Y_{j,m},$$

where $j = m - \dim(A_{m-1})$ and $m \geq 1$.

Theorem 4.1 ([25]). *Let $\{Z_i\}_{i=1}^r$ be a sequence of independent random self-adjoint $N \times N$ matrices. Suppose that $\mathbb{E}(Z_i) = 0$, $\|Z_i\| \leq B$ and let $\sigma^2 = \left\| \sum_{i=1}^r \mathbb{E}(Z_i^2) \right\|$. Then for all $t > 0$,*

$$\mathbb{P}\left(\lambda_{\max}\left(\sum_{i=1}^r Z_i\right) \geq t\right) \leq N \exp\left(-\frac{t^2/2}{\sigma^2 + Bt/3}\right), \tag{4.1}$$

where $\lambda_{\max}(U)$ is the largest singular value of a matrix U so that $\|U\| = \left(\lambda_{\max}(U^*U)\right)^{\frac{1}{2}}$ is the operator norm.

Let $N = \dim(A_{m_0+1})$ and $\mathcal{P}_N$ denotes the finite-dimensional subspace of $\mathcal{H}_\Psi$, spanned by $\{\phi_1, \ldots, \phi_N\}$. In particular, $\mathcal{P}_N = A_{m_0+1}$.

Let $\{x_i : i \in \mathbb{N}\}$ be random points i.i.d. uniformly distributed over $\mathbb{S}^{n-1}$ and define the random matrices Z_i as follows: For each $i \in \mathbb{N}$, consider the $N \times N$ random matrix M_i with entries

$$\begin{aligned} (M_i)_{lm} &= \sqrt{\lambda_l \lambda_m}\phi_l(x_i)\overline{\phi_m(x_i)}, \\ Z_i &= M_i - \mathbb{E}(M_i). \end{aligned}$$

Let $p \in \mathcal{P}_N$ with $\|p\|_{\mathcal{H}_\Psi} = 1$, then

$$p = \sum_{k=1}^N c_k \phi_k, \quad \text{and} \quad \sum_{k=1}^N |c_k|^2 \frac{1}{\lambda_k} = 1.$$

We denote $\tilde{c}_k = \frac{c_k}{\sqrt{\lambda_k}}$, then $\sum_{k=1}^N |\tilde{c}_k|^2 = 1$. Therefore, for each $i \in \mathbb{N}$

$$\begin{aligned} |p(x_i)|^2 &= \sum_{l,m=1}^N c_l \overline{c_m} \phi_l(x_i)\overline{\phi_m(x_i)} \\ &= \sum_{l,m=1}^N \tilde{c}_l \overline{\tilde{c}_m} \sqrt{\lambda_l \lambda_m} \phi_l(x_i)\overline{\phi_m(x_i)} \\ &= \langle \tilde{c}, M_i \tilde{c} \rangle. \end{aligned}$$

Hence, the expectation of $|p(x_i)|^2$ is given by

$$\begin{aligned}\frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}|p(x)|^2\,d\sigma_n(x) &= \frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}\sum_{l,m=1}^{N} c_l\overline{c_m}\phi_l(x)\overline{\phi_m(x)}\,d\sigma_n(x)\\ &= \sum_{l,m=1}^{N}\tilde{c}_l\overline{\tilde{c}_m}\frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}\sqrt{\lambda_l\lambda_m}\phi_l(x)\overline{\phi_m(x)}\,d\sigma_n(x)\\ &= \sum_{l,m=1}^{N}\tilde{c}_l\big(\mathbb{E}(M_i)\big)_{lm}\overline{\tilde{c}_m}\\ &= \langle\tilde{c},\mathbb{E}(M_i)\tilde{c}\rangle.\end{aligned}$$

Using the above observation, we obtain Lemma 4.2.

Lemma 4.2. *Let σ_n be the surface measure of $\mathbb{S}^{n-1}$. Let $X=\{x_i : i\in\mathbb{N}\}$ be an i.i.d. random variable uniformly distributed over $\mathbb{S}^{n-1}$. Then for $\gamma>0$, the event*

$$\mathcal{E}=\left\{\inf_{p\in\mathcal{P}_N,\,\|p\|_{\mathcal{H}_\Psi}=1}\sum_{i=1}^{r}\Big(|p(x_i)|^2-\frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}|p(x)|^2\,d\sigma_n(x)\Big)\geq-\gamma\right\}$$

and the event

$$\mathcal{F}=\left\{\lambda_{\max}\Big(\sum_{i=1}^{r}-Z_i\Big)\leq\gamma\right\}$$

are equivalent.

Proof. If U is an $N\times N$ self-adjoint matrix, then $\lambda_{\min}(U)=-\lambda_{\max}(-U)$, where $\lambda_{\min}(U)$ is the smallest singular value of U. Now,

$$\begin{aligned}&\inf_{p\in\mathcal{P}_N,\,\|p\|_{\mathcal{H}_\Psi}=1}\sum_{i=1}^{r}\Big(|p(x_i)|^2-\frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}|p(x)|^2\,d\sigma_n(x)\Big)\\ &=\inf_{\tilde{c}\in\mathbb{C}^N,\,\|\tilde{c}\|_2=1}\sum_{i=1}^{r}\Big(\langle\tilde{c},M_i\tilde{c}\rangle-\langle\tilde{c},\mathbb{E}(M_i)\tilde{c}\rangle\Big)\\ &=\inf_{\tilde{c}\in\mathbb{C}^N,\,\|\tilde{c}\|_2=1}\sum_{i=1}^{r}\langle\tilde{c},Z_i\tilde{c}\rangle\\ &=\lambda_{\min}\Big(\sum_{i=1}^{r}Z_i\Big).\end{aligned}$$

This completes the proof. $\square$

Proof of Theorem 1.1. The random matrix Z_i is a self-adjoint matrix with $\mathbb{E}(Z_i)=0$ for each $i\in\mathbb{N}$. To obtain matrix norm of Z_i, recall that for all $f\in\mathcal{H}_\Psi$,

$$\|f\|_{L^\infty(\mathbb{S}^{n-1})}\leq D\|f\|_{\mathcal{H}_\Psi}\quad\text{and}\quad\|f\|_{L^2(\mathbb{S}^{n-1})}\leq\|f\|_{\mathcal{H}_\Psi}.$$

Hence, we get

$$\begin{aligned}
\|Z_i\| &= \sup_{\tilde{c}\in\mathbb{C}^N,\, \|\tilde{c}\|_2=1} \left\{\langle \tilde{c}, Z_i\tilde{c}\rangle\right\} \\
&= \sup_{\tilde{c}\in\mathbb{C}^N,\, \|\tilde{c}\|_2=1} \left\{\langle \tilde{c}, M_i\tilde{c}\rangle - \langle \tilde{c}, \mathbb{E}(M_i)\tilde{c}\rangle\right\} \\
&= \sup_{p\in\mathcal{P}_N,\, \|p\|_{\mathcal{H}_\Psi}=1} \left\{|p(x_i)|^2 - \frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}} |p(x)|^2\, d\sigma_n(x)\right\} \\
&\le D^2.
\end{aligned}$$

To estimate $\mathbb{E}(Z_i^2)$, first we calculate $E(M_i)$ and $\mathbb{E}(M_i^2)$.

$$\begin{aligned}
(E(M_i))_{lm} &= \frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}} \sqrt{\lambda_l\lambda_m}\phi_l(x)\overline{\phi_m(x)}\, d\sigma_n(x) \\
&= \frac{1}{\omega_{n-1}}\sqrt{\lambda_l\lambda_m}\delta_{lm},
\end{aligned}$$

where δ_{lm} is the Kronecker delta. Inequality (2.3) implies

$$\begin{aligned}
(M_i^2)_{lm} &= \sum_{k=1}^{N}(M_i)_{lk}(M_i)_{km} \\
&= \sqrt{\lambda_l\lambda_m}\phi_l(x_i)\overline{\phi_m(x_i)}\sum_{k=1}^{N}\lambda_k|\phi_k(x_i)|^2 \\
&\le D^2\sqrt{\lambda_l\lambda_m}\phi_l(x_i)\overline{\phi_m(x_i)}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
(\mathbb{E}(M_i^2))_{lm} &\le \frac{D^2}{\omega_{n-1}}\sqrt{\lambda_l\lambda_m}\int_{\mathbb{S}^{n-1}} \phi_l(x)\overline{\phi_m(x)}\, d\sigma_n(x) \\
&= \frac{D^2}{\omega_{n-1}}\sqrt{\lambda_l\lambda_m}\delta_{lm}.
\end{aligned}$$

Let U and V be two square matrices of order N. The notation $U \le V$ stands for $(U)_{ij} \le (V)_{ij}$ for all $i, j = 1, \ldots, N$. Then we have $\mathbb{E}(M_i) \ge \frac{1}{\omega_{n-1}}D_N$, and $\mathbb{E}(M_i^2) \le \frac{D^2}{\omega_{n-1}}D_N$, where D_N is an $N\times N$ diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_N)$. Now,

$$\begin{aligned}
\mathbb{E}(Z_i^2) &= \mathbb{E}(M_i^2) - \mathbb{E}(M_i)^2 \\
&\le \frac{1}{\omega_{n-1}}\Big(D^2 - \frac{1}{\omega_{n-1}}\Big)D_N \le \frac{D^2}{\omega_{n-1}}D_N,
\end{aligned}$$

and $\sigma^2 = \Big\|\sum_{i=1}^{r}\mathbb{E}(Z_i^2)\Big\| \le \frac{rD^2}{\omega_{n-1}}$.

For $\gamma = r\nu$, Theorem 4.1 and Lemma 4.2 imply

$$\begin{aligned}\mathbb{P}(\mathcal{F}^c) &\leq \mathbb{P}\bigg(\lambda_{\max}\Big(\sum_{i=1}^{r} -Z_i\Big) \geq r\nu\bigg)\\ &\leq N \exp\bigg(-\frac{r^2\nu^2/2}{rD^2/\omega_{n-1} + D^2 r\nu/3}\bigg)\\ &\leq (n+1)(m_0+1)^{n-1}\exp\bigg(-\frac{\omega_{n-1}}{D^2}\frac{r\nu^2}{3+\omega_{n-1}\nu}\bigg)\\ &\leq 2^{n-1}(n+1)\bigg(1+\Big(\frac{2(n-2)!}{C}\Big)^{\frac{n-1}{n+\alpha-2}}\bigg)\exp\bigg(-\frac{\omega_{n-1}}{D^2}\frac{r\nu^2}{3+\omega_{n-1}\nu}\bigg).\end{aligned}$$

Hence, for all $p \in A_{m_0+1}$, the inequality

$$\frac{1}{r}\sum_{i=1}^{r}|p(x_i)|^2 - \frac{1}{\omega_{n-1}}\int_{\mathbb{S}^{n-1}}|p(x)|^2\,d\sigma_n(x) \geq -\nu\|p\|^2_{\mathcal{H}_\Psi}$$

holds with probability $\mathbb{P}(\mathcal{E}) = 1 - \mathbb{P}(\mathcal{F}^c)$.

Therefore, Lemma 3.3 implies the sampling inequality (1.2) holds for all $f \in \mathcal{H}_\Psi(\delta)$ with probability at least

$$1 - 2^{n-1}(n+1)\bigg(1+\Big(\frac{2(n-2)!}{C}\Big)^{\frac{n-1}{n+\alpha-2}}\bigg)\exp\bigg(-\frac{\omega_{n-1}}{D^2}\frac{r\nu^2}{3+\omega_{n-1}\nu}\bigg).$$

This completes the proof. □

Remark 4.3. The above result states that for given $\epsilon \in (0,1)$, the uniformly distributed random sampling set is a stable sample set for $\mathcal{H}_\Psi(\delta)$ with probability $1-\epsilon$ if

$$r > \frac{D^2(3+\nu\omega_{n-1})}{\nu^2\omega_{n-1}}\log\frac{2^{n-1}(n+1)\Big(1+\Big(\frac{2(n-2)!}{C}\Big)^{\frac{n-1}{n+\alpha-2}}\Big)}{\epsilon}$$

and $0 < \nu < \frac{1-2\sqrt{\delta}}{\omega_{n-1}} - 2D^2\sqrt{2\delta}$. This implies, if δ increases then the choice of ν decreases. For the explicit relation between δ and the sample size r, we consider $\nu = \eta\Big(\frac{1-2\sqrt{\delta}}{\omega_{n-1}} - 2D^2\sqrt{2\delta}\Big)$, where $\eta \in (0,1)$. Then the sampling inequality (1.2) holds for all $\mathcal{H}_\Psi(\delta)$ with probability $1-\epsilon$ if the number of samples

$$r > \frac{D^2\omega_{n-1}\Big(3+\eta\big(1-2\sqrt{\delta}-2D^2\omega_{n-1}\sqrt{2\delta}\big)\Big)}{\eta^2\big(1-2\sqrt{\delta}-2D^2\omega_{n-1}\sqrt{2\delta}\big)^2}\log\frac{2^{n-1}(n+1)\Big(1+\Big(\frac{2(n-2)!}{C}\Big)^{\frac{n-1}{n+\alpha-2}}\Big)}{\epsilon}.$$

4.1. **A general framework.** The above random sampling inequality can be proved in more general settings. Observe that the proofs of Theorem 1.1 and Lemma 3.3 hardly use any characteristic of spherical harmonic polynomials. In fact, we use only the following properties.

(a) The collection $\{Y_{j,k} : 1 \leq j \leq d_k, k \in \mathbb{N}_0\}$ of spherical harmonic polynomial is an orthonormal basis of $L^2(\mathbb{S}^{n-1})$.
(b) The sequence $\{\psi_k\}$ is a non-decreasing sequence of positive real with the rate of $d_k k^\alpha$, $\alpha > 1$. This fact guarantees that $\mathcal{H}_\Psi$ is RKHS and helps to calculate the probability estimation.

For the random sampling result in a general framework, we assume that M is a topological space with the probability measure μ on M. Furthermore, we assume

(i) $\{\phi_k : k \in \mathbb{N}\}$ is an orthonormal basis of $L^2(M, \mu)$.

(ii) $\Psi = \{\psi_k : k \in \mathbb{N}\}$ is a non-decreasing sequence with $\psi_1 = 1$ and $\psi_k > Ck^\alpha$, for some $C > 0$ and $\alpha > 0$.

(iii) There exists $D > 0$ such that

$$D^2 := \sup_{x\in M} \sum_{k\in\mathbb{N}} \frac{1}{\psi_k} |\phi_k(x)|^2 < \infty.$$

Hence the associated RKHS $\mathcal{H}_{\Psi,M}$ is defined by

$$\mathcal{H}_{\Psi,M} := \Big\{ f = \sum_{k\in\mathbb{N}} c_k\phi_k \in L^2(M, \mu) : \|f\|^2_{\mathcal{H}_{\Psi,M}} = \sum_{k\in\mathbb{N}} |c_k|^2\psi_k < \infty \Big\},$$

and the set of δ-concentrated functions in $\mathcal{H}_{\Psi,M}$ is

$$\mathcal{H}_{\Psi,M}(\delta) := \Big\{ f \in \mathcal{H}_{\Psi,M} : \|f\|^2_{L^2(M,\mu)} \geq (1-\delta)\|f\|^2_{\mathcal{H}_{\Psi,M}} \Big\}.$$

Theorem 4.4. *Let $\{x_j : j \in \mathbb{N}\}$ be a sequence of i.i.d. random variables and x_j is μ-distributed over M. Then for $\epsilon \in (0, 1)$, the sampling inequality*

$$\big(1 - 2\sqrt{\delta} - 2D^2\sqrt{2\delta} - \nu\big)\|f\|^2_{L^2(M,\mu)} \leq \frac{1}{r}\sum_{j=1}^{r} |f(x_j)|^2 \leq \frac{D_\alpha^2}{(1-\delta)}\|f\|^2_{L^2(M,\mu)}$$

holds for all $f \in \mathcal{H}_{\Psi,M}(\delta)$ with probability at least $1 - \epsilon$ if

$$r > \frac{D^2(3+\nu)}{\nu^2} \log \frac{2^{n+1}\big(1 + 2^{\frac{n-1}{\alpha}}\big)}{\epsilon}.$$

Here

$$\delta < \frac{1}{4(1 + D^2\sqrt{2})^2} \quad \textit{and} \quad 0 < \nu < 1 - 2\sqrt{\delta} - 2D^2\sqrt{2\delta}.$$

Proof. The result follows the line of proof of Theorem 1.1. The only minor modification occurs in Lemma 3.2, where the orthogonal projection P_m is defined as

$$P_m f = P_m\Big(\sum_{k\in\mathbb{N}} c_k\phi_k\Big) = \sum_{k\in\mathbb{N},\, \psi_k\leq m} c_k\phi_k.$$

The random variable $Y_j(f) = |f(x_j)|^2 - \|f\|^2_{L^2(M,\mu)} = |f(x_j)|^2 - \mathbb{E}(|f(x_j)|^2)$. □

Acknowledgment

We are grateful to the anonymous referee for meticulously reading the manuscript and giving us helpful suggestions.

References

[1] Akram Aldroubi and Karlheinz Gröchenig, *Nonuniform sampling and reconstruction in shift-invariant spaces*, SIAM Rev. **43** (2001), no. 4, 585–620, DOI 10.1137/S0036144501386986. MR1882684

[2] Douglas Azevedo and Valdir A. Menegatto, *Eigenvalues of dot-product kernels on the sphere*, Proceeding Series of the Brazilian Society of Computational and Applied Mathematics **3** (2015), no. 1, DOI 10.5540/03.2015.003.01.0039.

[3] Richard F. Bass and Karlheinz Gröchenig, *Random sampling of multivariate trigonometric polynomials*, SIAM J. Math. Anal. **36** (2004/05), no. 3, 773–795, DOI 10.1137/S0036141003432316. MR2111915

[4] Richard F. Bass and Karlheinz Gröchenig, *Random sampling of bandlimited functions*, Israel J. Math. **177** (2010), 1–28, DOI 10.1007/s11856-010-0036-7. MR2684411

[5] Richard F. Bass and Karlheinz Gröchenig, *Relevant sampling of band-limited functions*, Illinois J. Math. **57** (2013), no. 1, 43–58. MR3224560

[6] Albrecht Böttcher, Stefan Kunis, and Daniel Potts, *Probabilistic spherical Marcinkiewicz-Zygmund inequalities*, J. Approx. Theory **157** (2009), no. 2, 113–126, DOI 10.1016/j.jat.2008.07.006. MR2510921

[7] Emmanuel J. Candès, Justin Romberg, and Terence Tao, *Robust uncertainty principles: exact signal reconstruction from highly incomplete frequency information*, IEEE Trans. Inform. Theory **52** (2006), no. 2, 489–509, DOI 10.1109/TIT.2005.862083. MR2236170

[8] Stanley H. Chan, Todd Zickler, and Yue M. Lu, *Monte Carlo non-local means: random sampling for large-scale image filtering*, IEEE Trans. Image Process. **23** (2014), no. 8, 3711–3725, DOI 10.1109/TIP.2014.2327813. MR3234215

[9] Felipe Cucker and Lenore Blum, *The work of Steve Smale on the theory of computation: 1990–1999*, Foundations of computational mathematics (Hong Kong, 2000), World Sci. Publ., River Edge, NJ, 2002, pp. 15–33. MR2021976

[10] Feng Dai and Yuan Xu, *Approximation theory and harmonic analysis on spheres and balls*, Springer Monographs in Mathematics, Springer, New York, 2013, DOI 10.1007/978-1-4614-6660-4. MR3060033

[11] Matthieu Dolbeault, David Krieg, and Mario Ullrich, *A sharp upper bound for sampling numbers in L_2*, Appl. Comput. Harmon. Anal. **63** (2023), 113–134, DOI 10.1016/j.acha.2022.12.001. MR4525968

[12] Yonina C. Eldar, *Compressed sensing of analog signals in shift-invariant spaces*, IEEE Trans. Signal Process. **57** (2009), no. 8, 2986–2997, DOI 10.1109/TSP.2009.2020750. MR2723041

[13] Frank Filbir and W. Themistoclakis, *Polynomial approximation on the sphere using scattered data*, Math. Nachr. **281** (2008), no. 5, 650–668, DOI 10.1002/mana.200710633. MR2401509

[14] Willi Freeden, M. Zuhair Nashed, and Michael Schreiner, *Spherical sampling*, Geosystems Mathematics, Birkhäuser/Springer, Cham, 2018, DOI 10.1007/978-3-319-71458-5. MR3791815

[15] Hartmut Führ and Jun Xian, *Relevant sampling in finitely generated shift-invariant spaces*, J. Approx. Theory **240** (2019), 1–15, DOI 10.1016/j.jat.2018.09.009. MR3926050

[16] Karlheinz Gröchenig, *Irregular sampling, Toeplitz matrices, and the approximation of entire functions of exponential type*, Math. Comp. **68** (1999), no. 226, 749–765, DOI 10.1090/S0025-5718-99-01029-7. MR1613711

[17] Karlheinz Gröchenig, *Sampling, Marcinkiewicz–Zygmund inequalities, approximation, and quadrature rules*, J. Approx. Theory **257** (2020), 105455.

[18] Karlheinz Gröchenig and Joaquim Ortega-Cerdà, *Marcinkiewicz-Zygmund inequalities for polynomials in Bergman and Hardy spaces*, J. Geom. Anal. **31** (2021), no. 7, 7595–7619, DOI 10.1007/s12220-020-00599-5. MR4289270

[19] H. Groemer, *Geometric applications of Fourier series and spherical harmonics*, Encyclopedia of Mathematics and its Applications, vol. 61, Cambridge University Press, Cambridge, 1996, DOI 10.1017/CBO9780511530005. MR1412143

[20] Kamen Ivanov and Pencho Petrushev, *Irregular sampling of band-limited functions on the sphere*, Appl. Comput. Harmon. Anal. **37** (2014), no. 3, 545–562, DOI 10.1016/j.acha.2014.05.001. MR3256787

[21] Lutz Kämmerer, Tino Ullrich, and Toni Volkmer, *Worst-case recovery guarantees for least squares approximation using random samples*, Constr. Approx. **54** (2021), no. 2, 295–352, DOI 10.1007/s00365-021-09555-0. MR4321776

[22] David Krieg and Mario Ullrich, *Function values are enough for L_2-approximation: Part II*, J. Complexity **66** (2021), Paper No. 101569, 14, DOI 10.1016/j.jco.2021.101569. MR4292366

[23] Q. T. Le Gia and H. N. Mhaskar, *Localized linear polynomial operators and quadrature formulas on the sphere*, SIAM J. Numer. Anal. **47** (2008/09), no. 1, 440–466, DOI 10.1137/060678555. MR2475947

[24] Yaxu Li, Qiyu Sun, and Jun Xian, *Random sampling and reconstruction of concentrated signals in a reproducing kernel space*, Appl. Comput. Harmon. Anal. **54** (2021), 273–302, DOI 10.1016/j.acha.2021.03.006. MR4241985

[25] Lester Mackey, Michael I. Jordan, Richard Y. Chen, Brendan Farrell, and Joel A. Tropp, *Matrix concentration inequalities via the method of exchangeable pairs*, Ann. Probab. **42** (2014), no. 3, 906–945, DOI 10.1214/13-AOP892. MR3189061

[26] Jordi Marzo and Bharti Pridhnani, *Sufficient conditions for sampling and interpolation on the sphere*, Constr. Approx. **40** (2014), no. 2, 241–257, DOI 10.1007/s00365-014-9252-4. MR3254619

[27] H. N. Mhaskar, F. J. Narcowich, and J. D. Ward, *Spherical Marcinkiewicz-Zygmund inequalities and positive quadrature*, Math. Comp. **70** (2001), no. 235, 1113–1130, DOI 10.1090/S0025-5718-00-01240-0. MR1710640

[28] M. Zuhair Nashed and Qiyu Sun, *Sampling and reconstruction of signals in a reproducing kernel subspace of $L^p(\mathbb{R}^d)$*, J. Funct. Anal. **258** (2010), no. 7, 2422–2452, DOI 10.1016/j.jfa.2009.12.012. MR2584749

[29] Dhiraj Patel and Sivananthan Sampath, *Random sampling in reproducing kernel subspaces of $L^p(\mathbb{R}^n)$*, J. Math. Anal. Appl. **491** (2020), no. 1, 124270, 18, DOI 10.1016/j.jmaa.2020.124270. MR4110265

[30] Tomaso Poggio and Steve Smale, *The mathematics of learning: dealing with data*, Notices Amer. Math. Soc. **50** (2003), no. 5, 537–544. MR1968413

[31] Steve Smale and Ding-Xuan Zhou, *Shannon sampling and function reconstruction from point values*, Bull. Amer. Math. Soc. (N.S.) **41** (2004), no. 3, 279–305, DOI 10.1090/S0273-0979-04-01025-0. MR2058288

[32] Steve Smale and Ding-Xuan Zhou, *Shannon sampling. II. Connections to learning theory*, Appl. Comput. Harmon. Anal. **19** (2005), no. 3, 285–302, DOI 10.1016/j.acha.2005.03.001. MR2186447

[33] Elias M. Stein and Guido Weiss, *Introduction to Fourier analysis on Euclidean spaces*, Princeton Mathematical Series, No. 32, Princeton University Press, Princeton, N.J., 1971. MR0304972

[34] Ingo Steinwart and Andreas Christmann, *Support vector machines*, Information Science and Statistics, Springer, New York, 2008. MR2450103

[35] V. N. Temlyakov, *The Marcinkiewicz-type discretization theorems*, Constr. Approx. **48** (2018), no. 2, 337–369, DOI 10.1007/s00365-018-9446-2. MR3848042

[36] Pascal J. Thomas, *Sampling sets for Hardy spaces of the disk*, Proc. Amer. Math. Soc. **126** (1998), no. 10, 2927–2932, DOI 10.1090/S0002-9939-98-04411-6. MR1452830

[37] Mario Ullrich, *On the worst-case error of least squares algorithms for L_2-approximation with high probability*, J. Complexity **60** (2020), 101484, 6, DOI 10.1016/j.jco.2020.101484. MR4123742

[38] Jianbin Yang and Xinzhu Tao, *Random sampling and approximation of signals with bounded derivatives*, J. Inequal. Appl., posted on 2019, Paper No. 107, 14, DOI 10.1186/s13660-019-2059-x. MR3941165

[39] A. Zygmund, *Trigonometric series: Vols. I, II*, Cambridge University Press, London-New York, 1968. Second edition, reprinted with corrections and some additions. MR0236587

Department of Mathematics, Indian Institute of Technology Delhi, New Delhi-110016, India

Email address: `dpatel.iitd@gmail.com`

Department of Mathematics, Indian Institute of Technology Delhi, New Delhi-110016, India

Email address: `siva@maths.iitd.ac.in`